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Secure Distribution Venture Pvt. Ltd.

What is Operating Asset Funding ?

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EXECUTIVE SUMMARY & PRODUCT ESSENCE

Operating Asset Funding (OAF) is a specialized corporate finance and structured debt mechanism engineered to advance non-dilutive capital against the established operational cash flows, net revenue-generating capacity, and physical asset backing of brownfield, income-producing capital assets.

By targeting de-risked, post-Commercial Operation Date (COD) assets—such as renewable power parks, toll roads, commercial real estate, logistics hubs, industrial plants, and telecom networks—OAF enables corporate borrowers to compress interest spreads by 200 to 350 basis points, refinance high-cost construction debt, fund brownfield repowering/expansions, and optimize balance sheet return on equity (ROE) without diluting promoter control.

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Chapter 1: Product Overview & Core Philosophy of Operating Asset Funding

Operating Asset Funding (OAF) is a specialized corporate finance and structured debt architecture designed to provide long-term, non-dilutive liquidity against the established operational cash flows, net revenue-generating capacity, and physical collateral value of existing, operational capital assets. Unlike greenfield project finance—which carries severe construction, land acquisition, environmental clearance, and commissioning risks—Operating Asset Funding targets de-risked, brownfield assets that have already achieved Commercial Operation Date (COD) and demonstrated a clean, verifiable operational track record.

The foundational philosophy of Operating Asset Funding rests on asset de-risking and yield optimization. Once a capital asset achieves operational stability, its risk profile undergoes a fundamental transformation: construction uncertainty drops to zero, revenue visibility becomes predictable under long-term Power Purchase Agreements (PPAs), toll concessions, or commercial lease contracts, and operating margins stabilize. Consequently, senior lenders can re-underwrite the asset at significantly lower interest rate spreads (compressing borrowing costs from 12.0%–14.0% down to 8.5%–10.5% p.a.), extend debt door-to-door tenures to match remaining useful economic life (10 to 15 years), and increase debt sizing based on stable Debt Service Coverage Ratios (DSCR >= 1.35x).

Solving Balance Sheet Bottlenecks & Asset-Liability Mismatch (ALM)

In the Indian corporate banking ecosystem, enterprises frequently face severe Asset-Liability Mismatches (ALM) when capital-intensive operating assets are financed using short-term working capital lines (Cash Credits/Overdrafts) or 3-to-5 year medium-term loans. Short debt tenures force aggressive annual principal debt service, draining operational liquidity, starving maintenance budgets, and restricting corporate growth. Operating Asset Funding solves this systemic ALM bottleneck by refinancing short-term debt into long-term, self-amortizing facilities that mirror the asset's multi-year cash generation profile.

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Parameter	Operating Asset Funding	Greenfield Project Finance	Sale-and-Leaseback (SLB)	Working Capital (CC/OD)	Corporate Term Loan
Asset Stage	Brownfield (Post-COD)	Greenfield (Pre-COD)	Brownfield / Operational	Operational Cycle	Corporate Entity Level
Risk Profile	Low (Operational)	High (Construction/Delay)	Low to Medium	Low (Short-term)	Medium (Balance Sheet)
Debt Tenor	7 to 15 Years	10 to 20 Years	5 to 10 Years	12 Month Auto-Renewal	3 to 7 Years
Cost of Capital	8.75% - 10.75% p.a.	11.50% - 13.50% p.a.	9.50% - 12.00% p.a.	9.00% - 11.50% p.a.	10.00% - 12.50% p.a.
Collateral Base	Exclusive Asset Mortgage	Project Assets & Land	Asset Title Transfer	Current Assets (Inventory)	General Corporate Assets
Promoter Equity	No Equity Dilution	30% - 40% Equity Required	No Equity Dilution	N/A (Short-term)	No Equity Dilution

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Chapter 2: Structural Architecture, Taxonomy & Operational Mechanics

Operating Asset Funding encompasses a diverse taxonomy of structured debt and trade finance variants, each tailored to specific operational asset classes, cash flow profiles, and corporate objectives:

Taxonomy of Operating Asset Funding Variants

1. Operational Asset-Backed Term Loans (ABTL): Senior secured term loans extended directly to the asset-owning corporate entity or Special Purpose Vehicle (SPV), backed by a first exclusive mortgage over land, civil structures, and capital machinery, paired with an escrow charge on asset revenues.

2. Revenue-Linked & Cash Flow Securitization Facilities: Structured debt facilities where repayment is directly securitized against contracted future operational revenues (e.g., PPA receipts, toll collections, or commercial lease rentals) flowing into a ring-fenced Escrow Trust & Retention Account (TRA).

3. Equipment Repowering & Life-Extension Funding: Specialized capital expenditure facilities provided to operational assets to fund technology upgrades, efficiency overhauls, or capacity debottlenecking (e.g., replacing 1.5 MW wind turbines with 3.3 MW high-efficiency turbines) to extend useful economic life and boost EBITDA.

4. Refinancing & Debt Yield Optimization Facilities: Takeout debt structures designed to replace high-cost construction-phase loans with low-cost operational debt post-COD, compressing interest spreads and releasing excess cash flows to corporate sponsors.

5. Sale-and-Leaseback (SLB) & Operating Lease Funding: Transferring legal ownership or usage rights of operational capital equipment to an institutional lessor or NBFC while retaining full operational management, generating an immediate lump-sum cash payout.

Priority Escrow Trust & Retention Account (TRA) Cash Waterfall

To ensure bankruptcy-remoteness and absolute debt service priority, all gross cash receipts generated by the operating asset must flow directly into a designated Escrow TRA Account managed by a neutral Escrow Agent Bank. The cash waterfall governs monthly disbursements in strict hierarchical order:

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Waterfall Priority	Cash Allocation Tier	Operational & Structural Mechanism
Priority 1	Statutory Taxes & GST	Direct deduction of GST (SAC 9973/9971), Corporate Income Tax, and local property taxes.
Priority 2	Approved O&M Expenses	Release of contractually budgeted monthly Operation & Maintenance (O&M) expenses.
Priority 3	Senior Debt Interest Servicing	Monthly interest payment sweep to senior OAF term loan lenders.
Priority 4	Senior Debt Principal Amortization	Monthly principal installment sweep into the Debt Amortization Sub-Account.
Priority 5	Reserve Account Top-Up	Funding Debt Service Reserve Account (DSRA) and Major Maintenance Reserve (MMRA).
Priority 6	Unrestricted Sponsor Surplus	Release of remaining excess cash flow as dividends or corporate working capital.

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Chapter 3: Regulatory, Statutory, Tax & Accounting Framework in India

Operating Asset Funding operates at the intersection of Reserve Bank of India (RBI) credit directions, the Companies Act 2013, Indian tax statutes, and Indian Accounting Standards (Ind AS):

Reserve Bank of India (RBI) Prudential Directives

1. Income Recognition and Asset Classification (IRAC) Directions: Banks and Systemically Important NBFCs must enforce 90-day NPA provisioning rules. OAF facilities require continuous monitoring of TRA accounts to prevent payment delays.
2. Large Exposure Framework (LEF): Commercial banks are subject to single-counterparty exposure limits capped at 20% of Tier-1 capital (extendable to 25% for connected groups). Large operating asset refinancings frequently utilize multi-bank consortiums or syndicated loan structures.
3. Security Creation & Asset Valuation Norms: RBI mandates independent valuation reports from IBBI-registered valuers for all physical assets mortgaged under OAF facilities, enforcing minimum Fixed Asset Coverage Ratios (FACR >= 1.25x).

Companies Act 2013 Statutory Guardrails

1. Section 180(1)(a) Special Shareholder Resolutions: Selling, leasing, or mortgaging the whole or substantially the whole of an undertaking requires prior approval via a Special Resolution passed by corporate shareholders.
2. Section 180(1)(c) Borrowing Caps: Requires shareholder approval if total proposed borrowings exceed the aggregate of the company's paid-up share capital, free reserves, and securities premium.
3. Section 77 ROC Charge Registration: Mandatory registration of exclusive or pari-passu mortgage charges over operating assets with the Registrar of Companies (ROC Form CHG-1) within 30 days of security deed execution.

Direct & Indirect Tax Incentives (Income Tax Act 1961 & CGST Act 2017)

1. Section 32 Depreciation Shields: Legal asset owners are entitled to statutory depreciation deductions under the Block of Assets method (15% for general plant & machinery, 40% for renewable energy equipment), delivering a 25.17% corporate tax shield.
2. Section 36(1)(iii) & Section 37(1) Tax Deductibility: 100% of interest expenses incurred on OAF term debt and operational lease payments are fully deductible from gross business income as operational expenses.
3. Goods and Services Tax (GST) Eligibility: Operating asset transactions and lease services (SAC 9973 / 9971) attract 18% GST, which is 100% recoverable as Input Tax Credit (ITC) u/s 16 of the CGST Act 2017.

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Chapter 4: Common Eligibility Criteria & Institutional Underwriting Parameters

Institutional underwriters evaluate Operating Asset Funding proposals through a rigorous quantitative and qualitative credit framework focused on asset quality, cash flow durability, and off-taker creditworthiness:

Underwriting Parameter	Institutional Threshold	Credit Rationale & Risk Assessment
Asset Operational History	Min 12 to 36 Months Post-COD	Verifies actual vs projected production, plant load factor (PLF), and operating margins.
Debt Service Coverage (DSCR)	Average DSCR >= 1.35x - 1.60x	Ensures sufficient operational cash buffer to service senior debt principal and interest.
Fixed Asset Coverage (FACR)	FACR >= 1.25x - 1.75x	Provides first exclusive tangible asset collateral backing in event of liquidation.
Consolidated Leverage	Net Debt / EBITDA <= 3.5x - 4.5x	Caps total enterprise leverage to maintain long-term balance sheet solvency.
Off-taker Credit Quality	CRISIL / ICRA BBB+ to AAA	Mitigates counterparty payment default risk under PPAs, leases, or off-take contracts.
CIBIL Commercial Credit Rank	CMR 1 to CMR 4 Rank	Ensures impeccable historical debt repayment track record with zero prior default.

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Chapter 5: Security Architecture, Collateral Controls & Master Documentation Suite

Operating Asset Funding structures utilize a robust five-pillar security architecture designed to ring-fence operational cash flows and establish first-ranking legal recourse:

1. First Exclusive Legal Mortgage & Hypothecation: Registered legal mortgage over land, civil structures, plant, machinery, and equipment, perfected via ROC Form CHG-1 filing within 30 days u/s 77 Companies Act 2013.

2. Ring-Fenced Escrow TRA Cash Waterfall: Irrevocable assignment of all gross asset revenues into a designated Escrow TRA Account with automated priority cash sweeps.

3. Debt Service Reserve Account (DSRA): Cash Fixed Deposit or Demand Bank Guarantee covering 1 to 2 quarters of forward principal and interest debt service.

4. Legal Assignment of Contracts & Insurance: Assignment of all Off-take Contracts, PPAs, O&M Agreements, and Industrial All-Risk Insurance Policies naming senior lenders as loss payees.

5. Sponsor Guarantees & Share Pledge: Irrevocable corporate parent guarantees and pledge of 51% to 100% dematerialized SPV equity shares u/s 176 of the Indian Contract Act 1872.

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Chapter 6: Indicative Tenure, Pricing Structure & Financial Mechanics

Operating Asset Funding tenures and pricing models are dynamically calibrated based on asset class, remaining economic life, and off-taker credit standing:

1. Tenure & Amortization Mechanics: Door-to-door tenures range from 7 to 15 years, structured as fully amortizing monthly or quarterly installments matched to the remaining useful economic life (UEL) of the asset.

2. Pricing Formula & Spread Analysis: Pricing is structured at Benchmark Rate (Repo Rate / 10-Yr G-Sec @ ~6.50%) + Credit Risk Spread (175–300 bps) + Asset Quality Spread (50–100 bps) = 8.75% to 11.25% p.a. floating/fixed.

3. Post-COD Refinancing Spread Compression: By refinancing post-COD operational assets, corporate borrowers achieve a 200 to 350 basis point interest rate reduction compared to construction debt, generating massive interest savings.

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Chapter 7: Suitable Sectors, Asset Classes & Business Profiles

Operating Asset Funding is ideally suited for capital-intensive industries possessing established, cash-generating physical infrastructure:

- Renewable Energy Parks: Solar PV plants, wind farms, and hydro units operating under long-term SECI / NTPC / State Discom PPAs.

- Highways & Toll Expressways: Operational BOT-Toll and Hybrid Annuity Model (HAM) highway corridors with established traffic history.

- Commercial Real Estate & Warehousing: Grade-A office parks, retail malls, and automated logistics hubs under long-term lease agreements.

- Manufacturing & Heavy Industry: Cement, steel, chemical, and pharmaceutical manufacturing plants undergoing brownfield expansion.

- Telecom & Digital Infrastructure: Telecom towers, fiber optic networks, and hyperscale data centers with long-term tenant contracts.

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Chapter 8: Key Advantages and Limitations Structural Analysis Matrix

Key Advantages of Operating Asset Funding	Structural Limitations & Risk Factors
100% Brownfield Risk Elimination: Refinances assets post-COD, bypassing greenfield construction & execution delays.	Strict Escrow TRA Cash Lockbox: Traps operational revenue, restricting promoter access to excess cash flow until debt service is cleared.
200 - 350 bps Interest Cost Compression: Achieves significantly lower interest rates compared to construction debt.	Asset Obsolescence Risk: Rapid technological advances (e.g., solar module efficiency) may impact long-term asset terminal value.
Non-Dilutive Equity Preservation: Unlocks capital and funds brownfield expansion without diluting promoter equity.	Ind AS 116 Balance Sheet Debt Recognition: Capitalizes debt liabilities, increasing overall corporate balance sheet leverage.
100% Tax Deductibility: Interest expenses and lease payments are fully tax-deductible u/s 36(1)(iii) and Sec 37(1).	Upfront Structuring & Legal Expenses: Requires independent TEV audits, IBBI asset valuations, and ROC charge registration fees.
ALM Optimization: Aligns 10 to 15 year repayment tenures with useful economic life, boosting operational DSCR.	Off-taker Concentration Risk: Financial viability depends heavily on the creditworthiness of single PPA or lease counterparties.

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Chapter 9: Comprehensive Real-World Case Study (INR 350 Crore Operating Asset Refinancing)

To illustrate the transformational power of Operating Asset Funding, consider the real-world case of Vanguard Industrial Assets & Clean Energy Private Limited (VIACEPL), a mid-tier infrastructure corporate operating a 150 MW Solar PV Power Park and an advanced Industrial Manufacturing Facility in Western India valued at INR 500.00 Crores.

The Financial Challenge & Liquidity Bottleneck

VIACEPL had funded construction using INR 350.00 Crores of high-cost construction debt @ 12.50% p.a. with an aggressive 5-year repayment schedule. Following COD, annual debt service stood at an unbearable INR 98.20 Crores (INR 43.75 Cr interest + INR 54.45 Cr principal amortization) against annual operating EBITDA of INR 68.00 Crores. This resulted in a deficit Debt Service Coverage Ratio (DSCR) of 0.69x, threatening imminent NPA classification and operational collapse.

The Operating Asset Funding Refinancing Solution

VIACEPL executed an **INR 350.00 Crore 12-Year Operating Asset Refinancing & Debt Repowering Facility @ 8.75% p.a. fixed** with a consortium of private credit funds and institutional banks, supported by an Escrow TRA cash waterfall.

Financial Metric	Pre-OAF (Year 1)	OAF Year 1	OAF Year 3	OAF Year 6	OAF Year 12
Debt Principal Outstanding	INR 350.00 Cr	INR 350.00 Cr	INR 301.20 Cr	INR 208.50 Cr	INR 0.00 Cr
Senior Debt Coupon Rate	12.50% p.a.	8.75% p.a.	8.75% p.a.	8.75% p.a.	8.75% p.a.
Annual Operating EBITDA	INR 68.00 Cr	INR 72.00 Cr	INR 78.50 Cr	INR 86.00 Cr	INR 98.00 Cr
Annual Total Debt Service	INR 98.20 Cr	INR 46.80 Cr	INR 46.80 Cr	INR 46.80 Cr	INR 46.80 Cr

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Debt Service Coverage (DSCR)	0.69x (Deficit)	1.54x (Healthy)	1.68x	1.84x	2.09x
Net Cash Retained by Sponsor	-INR 30.20 Cr	+INR 25.20 Cr	+INR 31.70 Cr	+INR 39.20 Cr	+INR 51.20 Cr

Strategic Outcome: Operating Asset Funding compressed senior interest rates by 375 bps (saving INR 84.50 Crores in cumulative interest), expanded DSCR from 0.69x to 1.54x, released INR 25.20 Crores in annual surplus cash flow to VIACEPL, and boosted Return on Invested Capital (ROIC) from 11.2% to 23.5%.

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Chapter 10: Insolvency (IBC 2016) Integration & Legal Enforcement

Operating Asset Funding structures incorporate rigorous statutory enforcement mechanisms under the SARFAESI Act 2002 and the Insolvency and Bankruptcy Code (IBC) 2016:

1. Enforcement under SARFAESI Act 2002: Lenders holding registered ROC Form CHG-1 charges can issue 60-day demand notices u/s 13(2) and take physical possession of mortgaged operating assets u/s 13(4) without judicial intervention for public auction.

2. Insolvency Moratorium & Priority 1 CIRP Costs (IBC Sec 14): In the event of Corporate Insolvency Resolution Process (CIRP) initiation, leased operational assets occupied by the Corporate Debtor are protected under Section 14(1)(d). However, ongoing post-commencement asset usage fees and essential operating expenses must be paid in full as Priority 1 CIRP Costs.

3. Liquidation Waterfall Rankings (IBC Sec 53): Secured OAF lenders holding registered exclusive charges rank under Priority 2 (pari-passu with 24-month workmen's dues), ensuring primary claim over asset liquidation proceeds ahead of unsecured financial and operational creditors.

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Chapter 11: Rookie Mastery Examination (15 High-Yield Test Q&As)

Q1: What is the primary definition and core financial objective of Operating Asset Funding (OAF)?

A1: Operating Asset Funding is a structured debt mechanism that provides capital against the operational cash flows and physical asset value of de-risked, brownfield (post-COD) assets. Its core objective is to compress borrowing costs, refinance high-cost construction debt, optimize ALM, and fund brownfield expansions without diluting promoter equity.

Q2: How does Operating Asset Funding differ from Greenfield Project Finance?

A2: Greenfield project finance carries high construction, land acquisition, and execution risks, attracting high interest rates (11.5%-13.5% p.a.). OAF targets operational, post-COD assets with established cash flows, allowing 200 to 350 bps lower interest rates (8.75%-10.75% p.a.) and longer repayment tenures.

Q3: What is the primary function of an Escrow Trust & Retention Account (TRA) in OAF structures?

A3: An Escrow TRA account ring-fences gross asset revenues, enforcing an irrevocable, hierarchical cash waterfall that prioritizes statutory taxes, O&M expenses, senior debt service, and DSRA reserves before releasing surplus cash to promoters.

Q4: What are the key statutory requirements under Companies Act 2013 for mortgaging operational assets?

A4: Mortgaging operational assets requires shareholder approval via Special Resolution u/s 180(1)(a) and Section 180(1)(c), plus mandatory ROC Form CHG-1 charge registration within 30 days u/s 77.

Q5: What direct tax benefits does Operating Asset Funding offer under the Income Tax Act 1961?

A5: Asset owners receive statutory depreciation deductions u/s 32 (15% for general plant, 40% for renewables), while 100% of interest expenses u/s 36(1)(iii) and operational lease rentals u/s 37(1) are fully tax-deductible.

Q6: What is the typical Debt Service Coverage Ratio (DSCR) threshold required by underwriters for OAF?

A6: Institutional underwriters mandate a projected average DSCR >= 1.35x to 1.60x, with a minimum stress-scenario DSCR >= 1.20x.

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Q7: What is the Fixed Asset Coverage Ratio (FACR) requirement for OAF facilities?

A7: Lenders require a first exclusive FACR >= 1.25x to 1.75x tangible net asset collateral cover.

Q8: How does OAF resolve Asset-Liability Mismatches (ALM) on corporate balance sheets?

A8: It replaces short-term 3-to-5 year borrowings or working capital overdrafts with 7-to-15 year self-amortizing term debt matched to the remaining useful economic life of the operational asset.

Q9: What is the accounting treatment of leased operational assets under Ind AS 116?

A9: Ind AS 116 requires lessees to recognize Right-of-Use (ROU) Assets and corresponding Lease Liabilities on their balance sheets for all leases exceeding 12 months.

Q10: What is the role of a Debt Service Reserve Account (DSRA) in OAF security architecture?

A10: A DSRA holds 1 to 2 quarters of forward debt service (principal + interest) in cash FDs or bank guarantees to insulate senior lenders against temporary operational cash flow disruptions.

Q11: Which sectors are best suited for Operating Asset Funding in India?

A11: Renewable energy parks (solar/wind), highway concessions (BOT/HAM), commercial real estate & logistics hubs, heavy manufacturing plants, telecom infrastructure, and data centers.

Q12: How are OAF claims prioritized under Section 53 of the Insolvency and Bankruptcy Code (IBC) 2016?

A12: Secured OAF lenders holding registered ROC Form CHG-1 charges rank under Priority 2 of the liquidation waterfall, pari-passu with 24-month workmen's dues.

Q13: What happens to operational asset usage during CIRP under Section 14 of IBC 2016?

A13: Leased assets occupied by the Corporate Debtor cannot be repossessed during moratorium; however, ongoing post-commencement usage fees must be paid in full as Priority 1 CIRP Costs.

Q14: What is the impact of post-COD refinancing on a corporate borrower's interest expense?

A14: Post-COD refinancing compresses interest rate spreads by 200 to 350 basis points, dramatically reducing annual debt service and expanding corporate net cash flows.

Q15: What is the statutory time limit for registering a mortgage charge with the ROC u/s 77 of Companies Act 2013?

A15: ROC Form CHG-1 must be filed within 30 days of security deed execution (extendable up to 300 days with prescribed ROC fees).

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